Class 9: Drude Model: Successes and Limitations

As discussed in earlier classes, we wish to understand from first principles, why materials display the properties that they do. In particular we wish to explain material properties based on our understanding of how constituents of the material behave, how they interact with each other and with the surroundings.

Using the Drude model we have obtained predictions for the electronic conductivity and thermal conductivity of materials. They are given by the following expressions:

Electronic conductivity:

$$\sigma = \frac{N_f e^2 \tau}{m}$$

Thermal conductivity:

$$K = \frac{1}{3}\langle v^2 \rangle \tau C_V^e$$

With the expressions above, it has been possible to obtain an understanding of these two properties independently. The equations above match experimental data reasonably well, which is the most critical aspect of validation of any theory.

Having examined these two properties independently, let us now see if there is any interrelationship between these properties.

Several electrical appliances such as fans and motors use windings through which current flows in order to create electro magnets. More the current, more powerful is the electromagnet. At the same time current flowing through any metallic conductor generates heat due to resistive heating. Therefore metals chosen for preparing windings have to possess high electrical conductivity, and consequently low electrical resistivity. The metal of choice for windings for electromagnets is copper, because it is a good conductor of electricity.

One of the technologies that uses thermal conductivity as a central aspect of its operation is a heat exchanger. Heat exchangers extract heat from one location and pass the heat energy on to another location. They are used in a variety of systems such as air conditioners, in laptop computers etc. The material of choice for this technology is also copper since it is also a good conductor of heat.

Generalizing further, it is seen that materials which are good conductors of electricity are often also good conductors of heat.

In the Drude model for metallic systems, free electrons carry out the task of electrical conductivity as well as thermal conductivity. The model therefore creates a situation where factors impacting electrical conductivity also impact thermal conductivity which

results in the model being consistent with the observation that good conductors of electricity are also good conductors of heat.

Around the year 1850, Wiedemann and Franz experimentally investigated the relationship between electrical and thermal conductivity for several metals. They discovered that the ratio of the electrical to the thermal conductivity was a constant for several metals, at a given temperature. In particular, they found:

$$\frac{K}{\sigma T} = constant = 2.4 \times 10^{-8}\, J\Omega s^{-1} K^{-2}$$

Are the predictions of the Drude model consistent with the above experimental observation? Using the results of the Drude model, let us write and expression for $\frac{K}{\sigma T}$

$$\frac{K}{\sigma T} = \frac{\frac{1}{3}\langle v^2\rangle \tau C_V^e}{T N_f e^2 \tau / m}$$

Simplifying, we get:

$$\frac{K}{\sigma T} = \frac{m\langle v^2\rangle C_V^e}{3T N_f e^2}$$

The kinetic theory of gases, discussed in Class 6, gave us the following results:

$$\frac{1}{2} m\langle v^2\rangle = \frac{3}{2} k_b T$$

$$C_V^e = \frac{3}{2} N_f k_b$$

Assuming these results hold for free electrons as well, by substituting these results in the equation above we get:

$$\frac{K}{\sigma T} = \frac{3}{2}\frac{k_b^2}{e^2}$$

This is then the prediction made by the Drude model. Substituting the values for the Boltzmann's constant, and the charge of an electron, we get

$$\frac{K}{\sigma T} = 1.1 \times 10^{-8}\, J\Omega s^{-1} K^{-2}$$

This is very close to the value obtained experimentally by Wiedemann and Franz, and certainly in the correct order of magnitude.

Correctly predicting the Wiedemann –Franz law, in addition to correctly predicting the electrical and thermal conductivities independently, are the major successes of the Drude model.

As indicated earlier, the Drude model has extended ideal gas laws to constituents of a solid, where the number density of particles is higher by three orders of magnitude. Therefore there is reason for concern. However, as we have just seen, despite such concerns, the model displays significant success in the predictions it makes.

As it turns out, the correct prediction of the thermal conductivity has occurred fortuitously.

The value of C_V^e, predicted using the ideal gas laws, is higher, by two orders of magnitude, than the experimental values obtained using low temperature measurements - where the electronic contributions are significant. In the next class we will see that we can predict the value of τ with reasonable confidence. Therefore the correct prediction of thermal conductivity implies that the prediction of $\langle v^2 \rangle$ is correspondingly lower by two orders of magnitude.

In addition to thermal and electrical conductivity, the Drude model also enjoys some success in predicting the Hall coefficient.

The Hall effect was discovered around the year 1880. A schematic of the effect is shown in Figure 9.1 below.

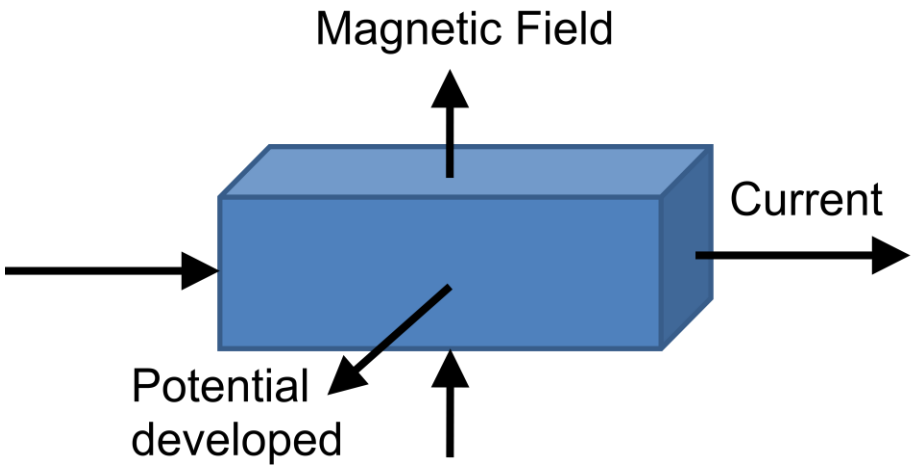


Figure 9.1: A schematic showing the appearance of the Hall effect in a conductor carrying current, which is subject to a magnetic field perpendicular to the current. A potential is developed which is perpendicular to the magnetic field as well as the current

It is an important effect in that it enables us to determine the sign of the charge carrier in a conductor. Measuring a current alone does not tell us anything about the sign of the

charge carrier in a conductor. It was noticed that if a magnetic field is placed perpendicular to the direction of a flowing current, the magnetic field deflects the charge carriers in a direction perpendicular to the magnetic field as well as the flowing current. A potential therefore develops perpendicular to the direction of flow of current. Build up of charge occurs till the potential developed opposes any further movement of charge in the perpendicular direction. Depending on the sign of the charge carrier, the potential is either positive or negative. The Hall coefficient, R_H, which results in the associated calculations, is negative if the charge carrier is negative, and is positive if the charge carrier is positive.

The Drude model is consistent with a negative R_H, but is not able to predict a positive value for R_H.

While a vast majority of the elements in the periodic table are metallic in nature, any general theory for materials should also account for semiconductors and insulators. While the Drude model does use N_f to distinguish between materials, this alone does not capture the differences between materials comprehensively. For example, the changes in material properties with changes in crystal structure, and the existence of anisotropy in a most crystalline solids, cannot be explained simply on the basis of N_f. N_f is the same regardless of direction. On the other hand, in an ideal gas there is no preferred orientation, which is the reason we have:

$$\langle v_x^2 \rangle = \frac{1}{3}\langle v^2 \rangle$$

In a crystalline solid there is distinct directionality in that the ionic cores are not randomly distributed. Therefore, to the extent that ionic cores impact material properties, the properties will also display directionality or anisotropy. In the Drude model we have largely ignored the presence of the ionic cores except to introduce a general resistive term γv, and it is therefore not surprising that the predictions demonstrate limitations. Therefore, the model needs o be further refined.

In summary, the Drude model successfully predicts electrical and thermal conductivity of metallic systems, and the Wiedemann Franz law, but makes incorrect predictions of C_V^e, $\langle v^2 \rangle$, and R_H. It is now of interest to see how we can improve the model. In particular, we need to identify the specific fundamental assumptions of the Drude model that need to be changed, and to identify the appropriate manner to incorporate these changes.